

GEORGIA INSTITUTE OF TECHNOLOGY
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
ECE 3040 Final Practice Test Solutions

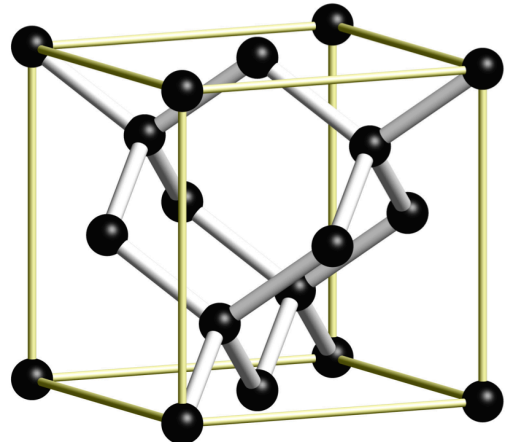
Name: Solutions

Contents: **15 Questions (50 points)**
3 Problems (60 points)

1. **(3 points)** For the lattice structure shown on the right

(a) (1 point) What type of lattice is this? Give an example material for this lattice structure?

Diamond lattice
Silicon, Germanium



(b) (1 point) What is the number of nearest neighbors for the atoms in this crystal lattice?

4

(c) (1 point) How many atoms are there in the unit cell?

8

2. (3 points) What is the probability of an energy state with energy $E_F + 2kT$ being occupied by an electron? What is the probability of a state with the same energy being occupied by a hole?

$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}} \Rightarrow f(E_F + 2kT) = \frac{1}{1 + e^2}$$

Hole occupation probability: $1 - \underbrace{f(E_F + 2kT)}_{0.88} \approx \boxed{0.12}$

3. (3 points) Consider three intrinsic semiconductor materials with the same effective density of conduction and valence band states. The bandgaps of these semiconductors are such that $E_{G1} > E_{G2} > E_{G3}$. Please qualitatively compare the equilibrium concentration of carriers (n_i) in these semiconductors,

(a) (1 point) at room temperature.

$$n_{i1} < n_{i2} < n_{i3}$$

(b) (1 point) at 0K

$$n_{i1} = n_{i2} = n_{i3}$$

(c) (1 point) at 600K

$$n_{i1} < n_{i2} < n_{i3}$$

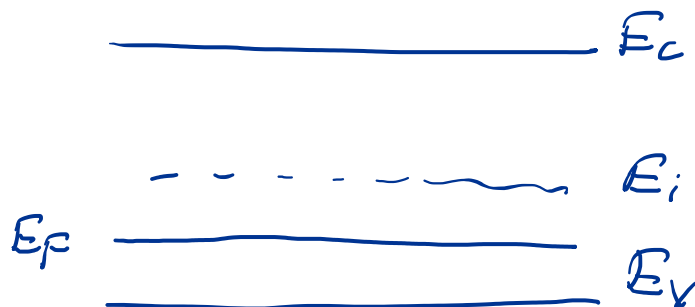
4. (3 points) A silicon wafer is simultaneously doped with both 10^{15} cm^{-3} boron and 10^{12} cm^{-3} phosphorus atoms.

- (a) (1.5 points) Calculate the electron and hole carrier densities at room temperature, assuming complete ionization ($n_i = 10^{10} \text{ cm}^{-3}$).

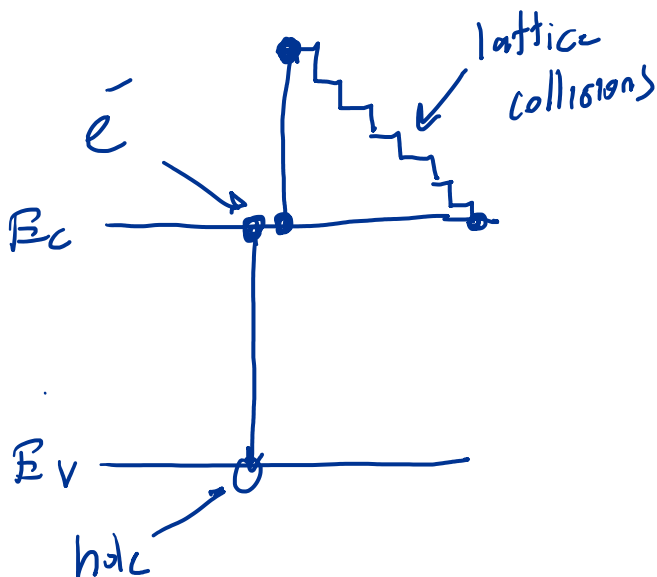
$$N_A \gg N_D \quad p = 10^{15} / \text{cm}^3$$

$$\rightarrow n = n_i^2 / p = 10^5 / \text{cm}^3$$

- (b) (1.5 points) Roughly sketch (no calculations necessary) the energy band diagram, properly showing the locations of the Fermi energy, intrinsic Fermi energy, conduction and valence band energy levels.



5. (3 points) Describe the Auger recombination process in words. Also, please draw a schematic showing the process on the energy-band diagram to support your description.



When a hole and an electron recombine, the energy is transferred to another electron as kinetic energy. This kinetic energy is then released as thermal energy with lattice collisions.

6. (3 points) Imagine an n^+p junction. Which current component (electron or hole, drift or diffusion) is dominating the current flow through the junction under following conditions? Justify your answer.

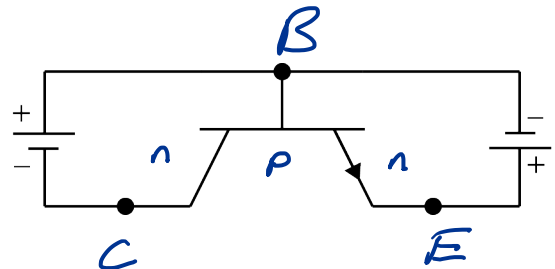
(a) (1.5 points) Under forward bias.

Under forward bias, majority carrier diffusion dominates. Since e^- concentration in n^+ side is higher than hole concentration in the p -side, the dominant component is electron diffusion current.

(b) (1.5 points) Under reverse bias.

Under reverse bias, minority carrier drift dominates. Since e^- concentration in p -side is higher than holes in the n^+ side, electron drift current dominates.

7. (3 points) Identify the type of BJT and mark the emitter, base and collector terminals. Assuming $N_E \gg N_B \gg N_C$, sketch the energy band diagram. In your sketch clearly indicate emitter, base and collector regions and label E_C , E_V , E_i and E_F .

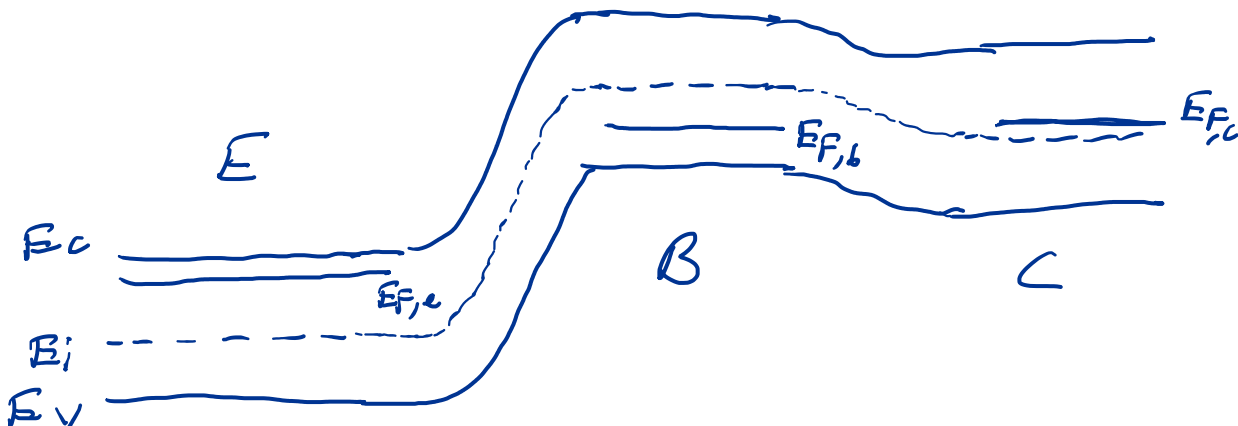


NPN

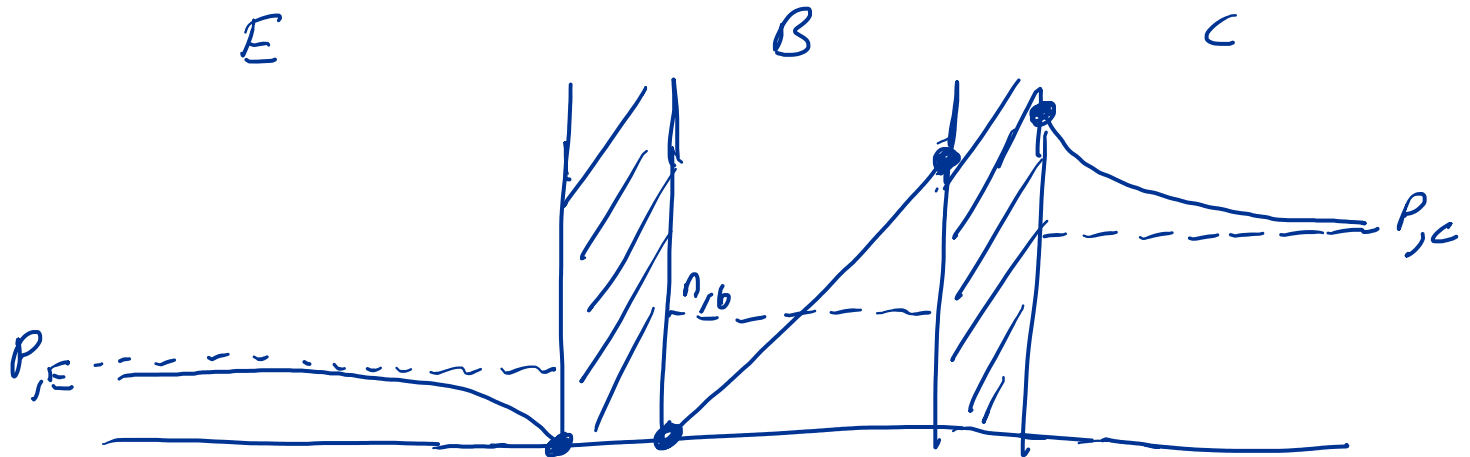
EB $\rightarrow$ reverse-biased

BC $\rightarrow$ forward-biased

$\Rightarrow$ Inverted Mode



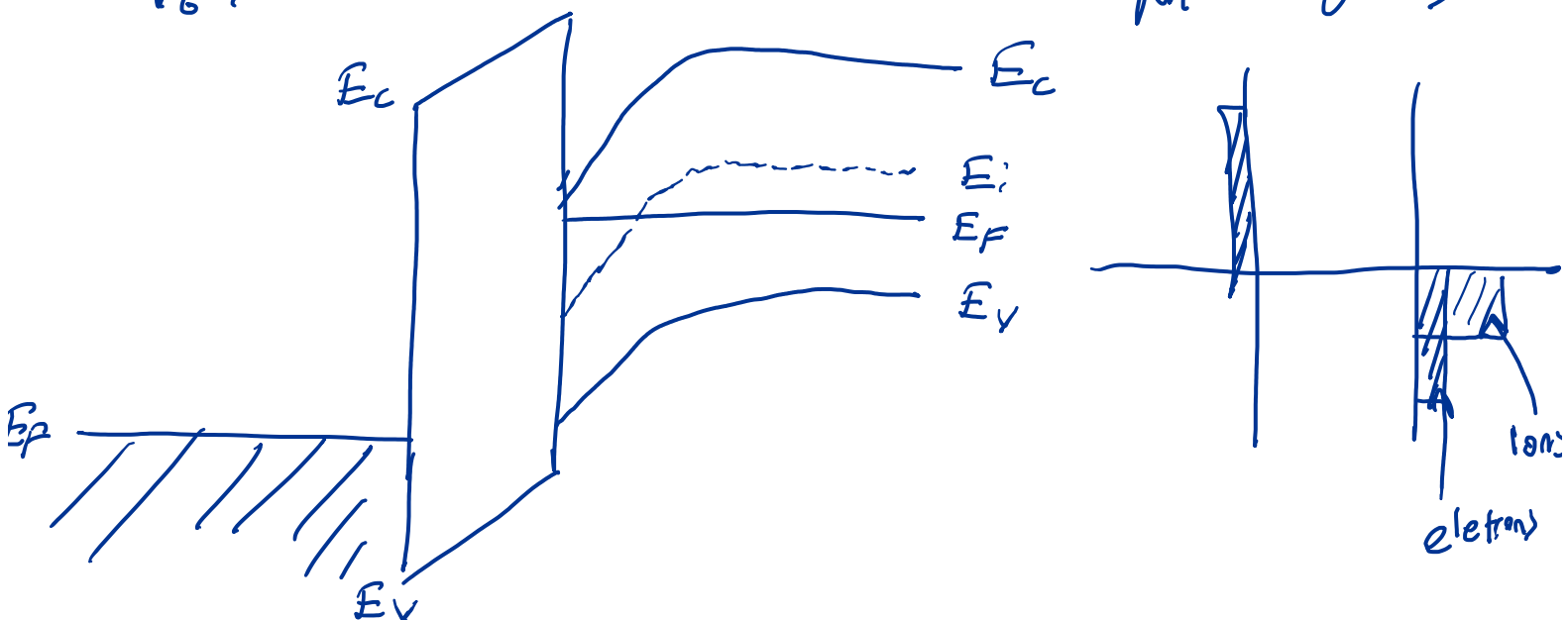
8. (3 points) For the BJT in question 7, roughly sketch the **total minority carrier** concentration in the emitter, base and collector regions. Please indicate the minority carrier type and equilibrium concentrations, mark the emitter, base and collector in your sketch, and assume $W/L_B \ll 1$.



9. (6 points) Sketch the **energy band diagram** and **charge diagram** of an ideal MOS (assume $\Phi_M = \Phi_S$) structure for the following conditions. In your energy band diagram, include E_F in the metal, E_C and E_V in the oxide, and E_C , E_i , E_F and E_V in the semiconductor. Also indicate the polarity of the potential difference between the metal and the semiconductor for each case (i.e., $V_G > 0$ or $V_G < 0$).

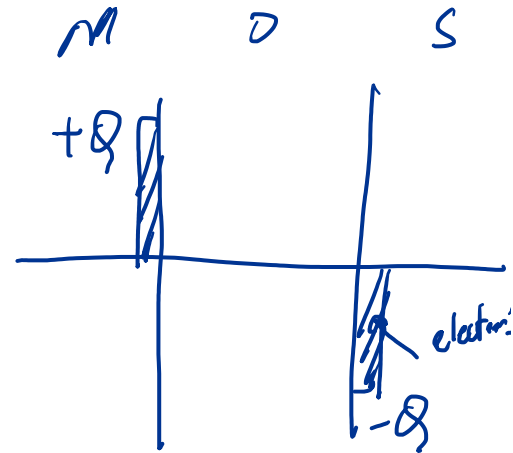
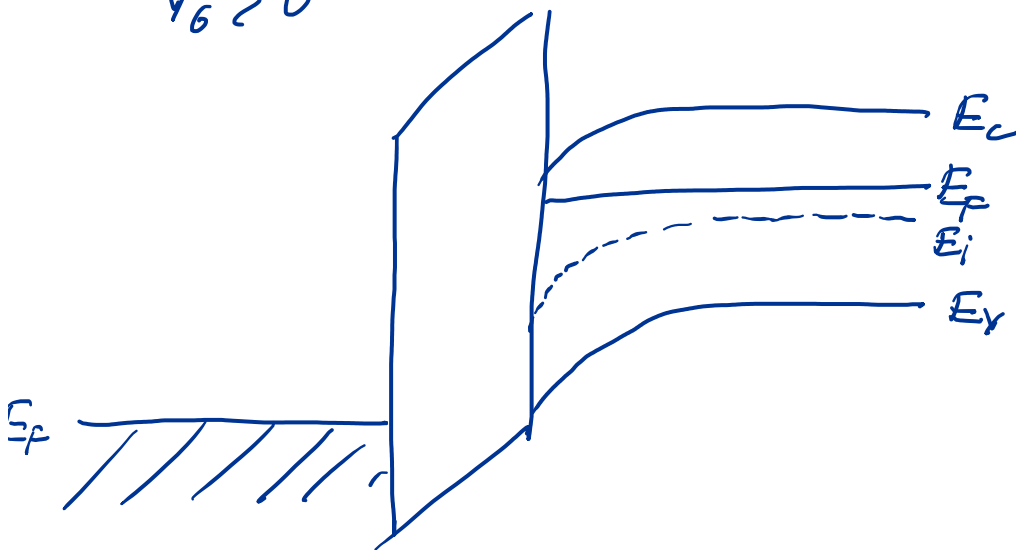
- (a) (3 points) p-type semiconductor **under inversion**.

$$V_G > 0$$



(b) (3 points) n-type semiconductor **under accumulation**.

$$V_G > 0$$



10. (3 points) Consider an MOS capacitor with a p-type Si substrate doped with 10^{17} cm^{-3} acceptor atoms at room temperature. Under delta depletion approximation, calculate the potential difference across the semiconductor

(a) (1 point) For accumulation biasing.

$$\phi_s = 0$$

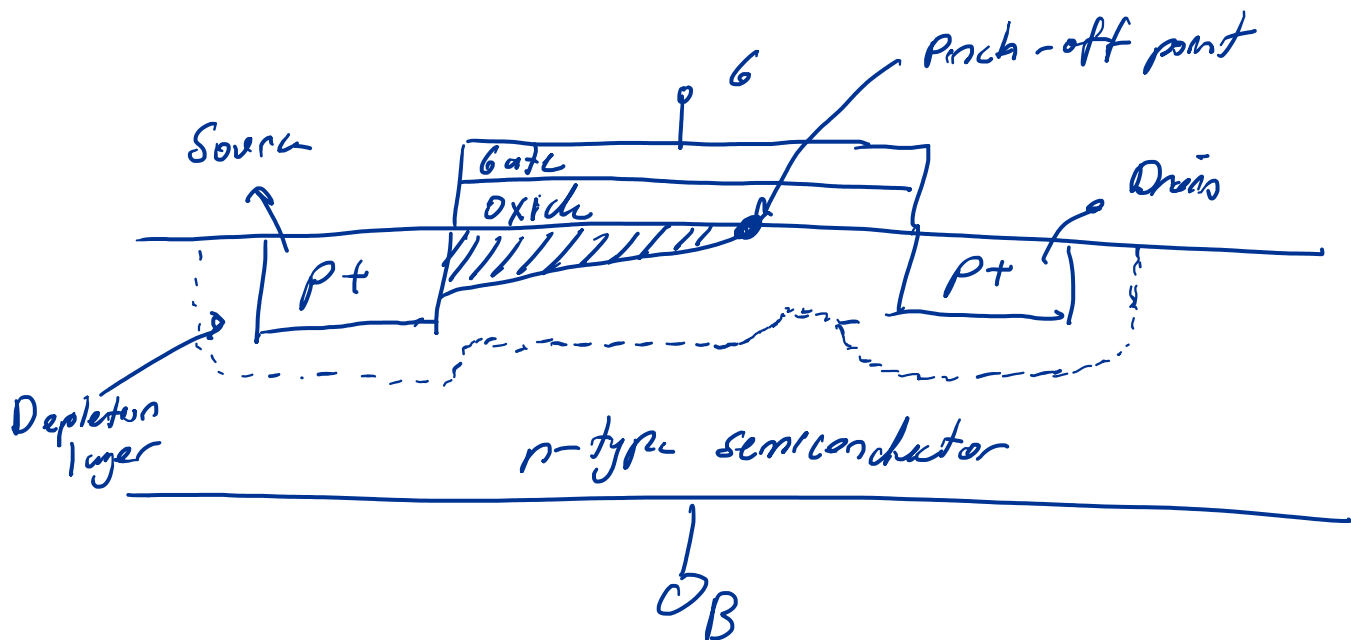
(b) (1 point) At the onset of inversion.

$$\phi_s = 2 \phi_F = 0.8381 \text{ V}$$

(c) (1 point) For inversion biasing.

$$\phi_s = 2 \phi_F = 0.8381 \text{ V}$$

11. (3 points) Sketch a cross-section of a **p-channel MOSFET** beyond the pinch-off point (in **saturation**), properly labeling the key regions (and their doping type) and the terminals. Include the depletion region and the inversion layer in your sketch.



12. (3 points) How are the I_D - V_{DS} characteristics of a **short channel** and **long channel** MOSFET different? Please indicate the underlying physical reason. Which parameter of the MOSFET's small signal equivalent circuit is affected by this difference?

Beyond pinch-off, the conductive channel becomes shorter. For long-channel device, the change in the channel length is negligible. For short channel device, the change in the channel length leads to decrease in the channel resistance and I_D increases with V_{DS} . This effect is represented by an output resistance (r_o) in the small signal model. r_o is affected by this modulation.

13. (5 points) Which of the following statements are true and which are false?

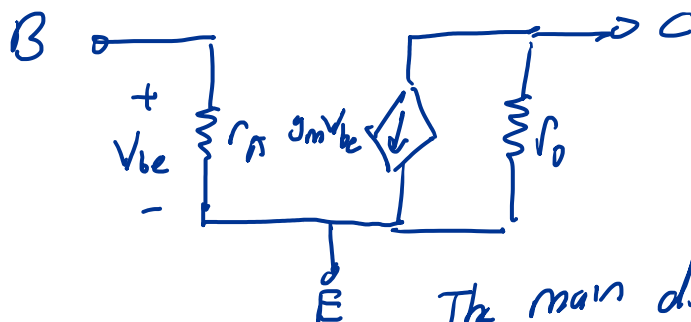
- (a) The electric field inside the gate insulator in an ideal MOSFET is constant. *True*
- (b) There is a current flow between the gate and the source in NMOS when biased in accumulation. *False*
- (c) A p-doped Si substrate has greater work function compared to an n-doped Si substrate. *True*
- (d) A PMOS has n⁺-type source and drain and a p-type substrate. *False*
- (e) In a PMOS, the drain has higher potential than the source. *False*

14. (3 points) Assume you want to build a single transistor amplifier using a MOSFET. Which terminals can you use for input? Which terminals can you use for the output? Justify your answer.

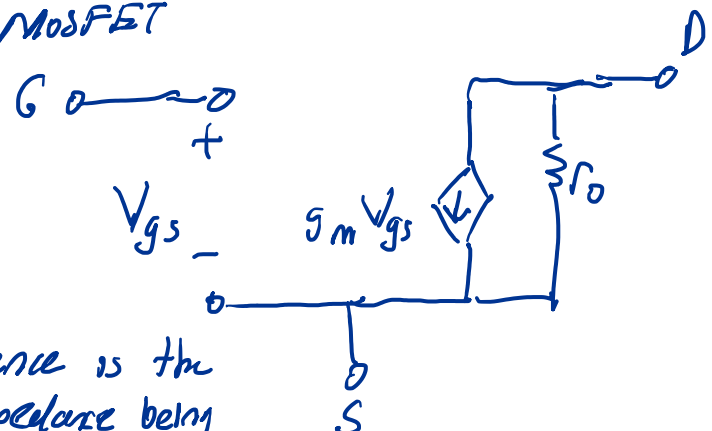
*Source and gate can be used for input
 Source and drain can be used for output.
 MOSFET current is mainly controlled by V_{GS} . So, gate or source terminal can be used to control the drain current. The drain current can be collected from drain or source terminals. Gate current is zero so it cannot be used for output.*

15. (3 points) Draw small signal models of both BJT (active-mode) and MOSFET (saturation). Include the output resistances. Clearly indicate the terminals. What is the main difference between them?

BJT (npn)

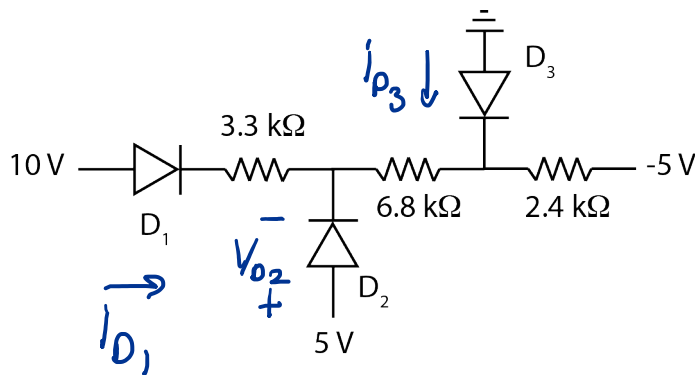


MOSFET



The main difference is the MOSFET input impedance being infinite

Problem 1 (10 points): Find the states (forward or reverse biased) and Q-points for the diodes in the circuit using the constant voltage drop model with $V_{on} = 0.7\text{ V}$.



Assume D_1 ON, D_2 OFF, D_3 ON

$$(1) \quad 10\text{V} - 0.7\text{V} - 3.3\text{k} i_{D1} - 6.8\text{k} i_{D1} = -0.7\text{V}$$

$$10\text{V} = 10.1\text{k} i_{D1} \Rightarrow i_{D1} = 0.99\text{mA}$$

$$(2) \quad -0.7 - 2.4\text{k} (i_{D3} + i_{D1}) = -5\text{V}$$

$$4.3\text{V} = 2.4\text{k} (i_{D3} + i_{D1}) \Rightarrow i_{D3} + i_{D1} = 1.79\text{mA}$$

$$\Rightarrow i_{D3} = 0.8\text{mA}$$

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$$-V_{D_2} = 10V - 0.7V - 3.3k(0.99mA) - 5V$$

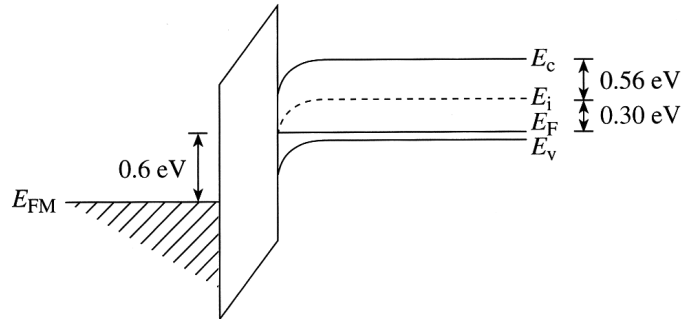
$$\Rightarrow V_{D_2} = -1.03V$$

$\Rightarrow$ 8 points : D_1 ON (0.7V, 0.99mA)

D_2 OFF (-1.03V, 0)

D_3 ON (0.7V, 0.8mA)

Problem 2 (25 points): Shown below is a energy band diagram for an ideal silicon MOS capacitor (i.e. the metal workfunction equals the semiconductor workfunction) operated at $T = 300$ K with $V_G \neq 0$. Note that $E_F = E_i$ at the Si-SiO₂ interface.



- (2 points) Extract the Fermi potential ϕ_F in [V] and the energy difference $E_F - E_{i,bulk}$ in [eV] far away from the Si-SiO₂ interface.
- (4 points) Calculate the carrier concentrations n and p in the quasineutral regime and determine the doping concentration of the substrate, assuming a single dopant type.
- (4 points) Calculate the resistivity of the semiconductor substrate far away from the Si-SiO₂ interface. Assume $\mu_n = 1360$ cm²/Vs and $\mu_p = 460$ cm²/Vs.
- (3 points) Extract the surface potential ϕ_s .
- (4 points) Extract the applied gate voltage V_G (including its sign)?
- (4 points) Calculate the oxide thickness x_o .
- (4 points) Calculate the threshold voltage V_T , i.e. the gate voltage required to reach the onset of inversion.

You might need: $K_O = 3.9$, $K_S = 11.7$, $\epsilon_0 = 8.85 \cdot 10^{-14}$ F/cm, $q = 1.6 \cdot 10^{-19}$ C, $kT = 0.026$ eV at room temperature, $E_g = 1.12$ eV, $n_i = 10^{10}$ cm⁻³.

$$(a) \quad \phi_F = \frac{1}{q} (E_{i,bulk} - E_F) = \boxed{0.3 \text{ V}}$$

$$E_F - E_{i,bulk} = \boxed{-0.3 \text{ eV}}$$

$$(b) \quad p = n_i e^{(E_F - E_i)/kT} = \boxed{1.02 \times 10^{15} / \text{cm}^3}$$

$$n = \frac{n_i^2}{p} = \boxed{9.75 \times 10^4 / \text{cm}^3}$$

$$N_A \approx p = \boxed{1.02 \times 10^{15} / \text{cm}^3} \quad \text{since } p \gg n$$

$$(c) \quad p\text{-type}, \quad p \gg n$$

$$\rho = \frac{1}{q \mu_p p} = \boxed{13.32 \, \Omega \text{cm}}$$

$$(d) \quad \phi_s = \frac{1}{q} (E_{i, \text{bulk}} - E_{i, \text{surface}}) = \boxed{0.3 \text{ V}}$$

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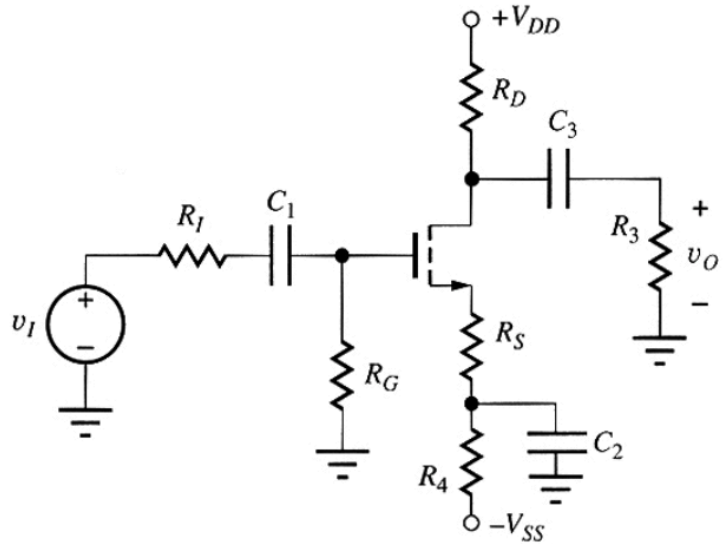
$$(e) \quad V_g = \frac{1}{q} (E_{F_{\text{semiconductor}}} - E_{F_{\text{metal}}}) = \boxed{0.6 \text{ V}}$$

$$(f) \quad V_g = \phi_s + \frac{K_s X_o}{K_o} \sqrt{\frac{2q N_A}{K_s \epsilon_o}} \phi_s$$
$$\Rightarrow X_o = \boxed{103 \text{ nm}}$$

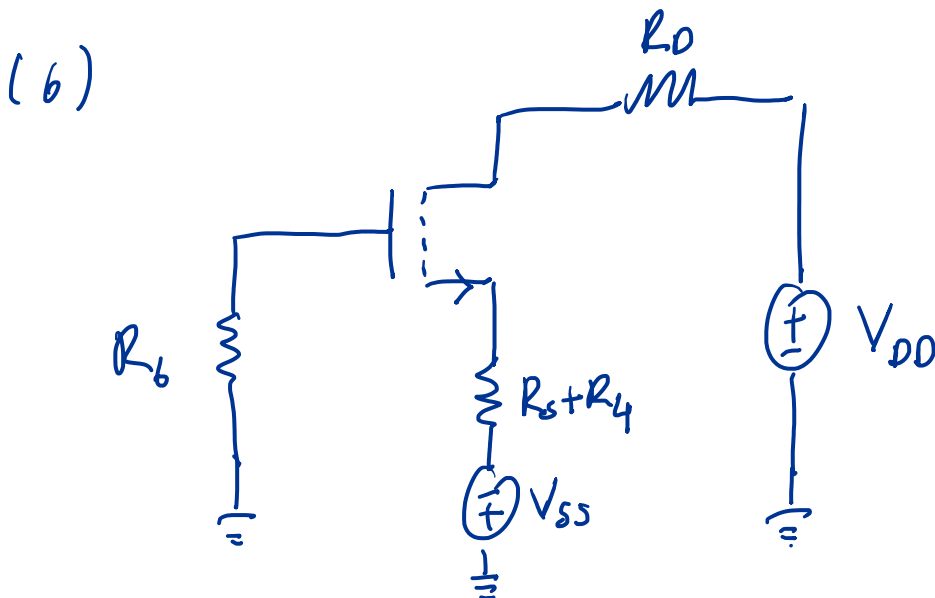
$$(g) \quad V_T = 2\phi_F + \frac{K_s X_o}{K_o} \sqrt{\frac{2q N_A}{K_s \epsilon_o}} 2\phi_F$$
$$= \boxed{1.025 \text{ V}}$$

Problem 3 (25 points): You are investigating the amplifier shown below with a MOSFET transistor operated in saturation mode bias. $R_I = 2 \text{ k}\Omega$, $R_G = 500 \text{ k}\Omega$, $R_D = 6 \text{ k}\Omega$, $R_S = 2 \text{ k}\Omega$, $R_4 = 3 \text{ k}\Omega$ and $R_3 = 15 \text{ k}\Omega$, $V_{DD} = 5 \text{ V}$, $V_{SS} = 5 \text{ V}$, $V_T = 2 \text{ V}$, $K_n = 600 \mu\text{A}/\text{V}^2$ and $\lambda = 0$.

- (1 point) Which single-transistor amplifier type is represented by this circuit?
- (4 points) Draw the dc equivalent circuit of the amplifier.
- (6 points) Calculate the Q-point (I_D , V_{DS}). Verify that the MOSFET is indeed saturation mode biased.
- (3 points) Calculate the transconductance g_m , the input resistance r_π and the output resistance r_o of the MOSFET.
- (5 points) Draw the ac equivalent circuit of the amplifier including the small signal equivalent circuit for the MOSFET.
- (6 points) Using the small signal equivalent circuit, extract the signal source voltage gain v_o/v_i of the amplifier.



(a) Common - source amplifier



$$(c) \quad V_G = 0, \quad V_{GS} = 0 - (I_D (R_S + R_H) - V_{SS}) = V_{SS} - I_D (R_S + R_H) \quad (1)$$

$$-V_{DD} + I_D R_D + V_{DS} + I_D (R_S + R_H) - V_{SS} = 0$$

$$V_{DS} = V_{DD} + V_{SS} - I_D (R_D + R_S + R_H) \quad (2)$$

$$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 \quad (3)$$

Using (1) and (3)

$$\begin{aligned} V_{GS} &= V_{SS} - (R_S + R_H) \frac{K_n}{2} (V_{GS} - V_T)^2 \\ &= 5 - 5 \times 10^3 \frac{600 \times 10^{-6}}{2} (V_{GS} - 2)^2 \\ &= 1.5 V_{GS}^2 - 5 V_{GS} + 1 = 0 \end{aligned}$$

$$\boxed{V_{GS} = 3.12V} \quad \cancel{V_{GS} = 0.21V}$$

$$\boxed{I_D = 0.376 \text{ mA}}$$

$$\boxed{V_{DS} = 5.86V}$$

$$, V_{DS} \geq V_{GS} - V_T \quad \checkmark$$

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$$(d) \quad g_m = \frac{2.10}{V_{GS} - V_T}$$

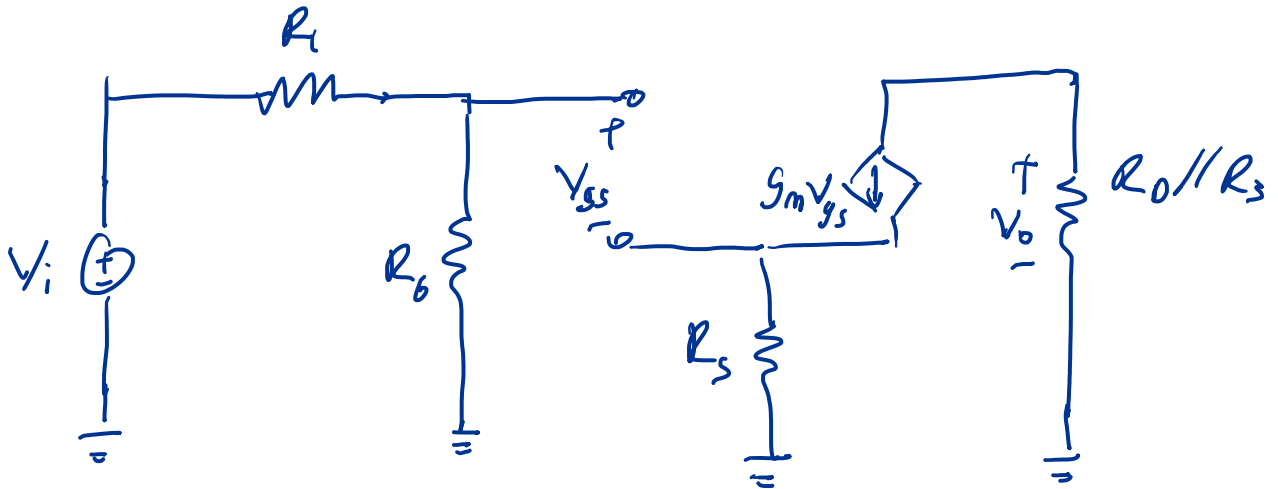
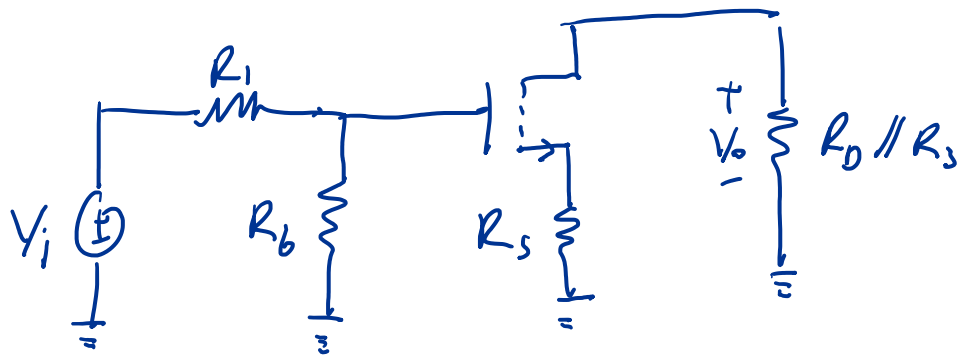
$$g_m = \frac{2 \times 0.376 \times 10^{-3}}{3.12 - 2}$$

$$g_m = \boxed{0.671 \text{ mS}}$$

$$r_{\pi} = \infty$$

$$r_o = \infty \quad (x = 0)$$

(e)



$$(f) \quad V_o = - (R_D \parallel R_S) g_m V_{gs}$$

$$V_g = \frac{R_6}{R_1 + R_6} V_I = 0.996 V_I$$

$$V_{gs} = 0.996 V_I - R_S g_m V_{gs}$$

$$V_I = \frac{V_{gs} (1 + g_m R_S)}{0.996}$$

$$\frac{V_o}{V_I} = \frac{- (R_D \parallel R_S) g_m \cancel{V_{gs}}}{\frac{(1 + g_m R_S) \cancel{V_{gs}}}{0.996}}$$

$$= \frac{- 0.996 \times 4.3 \times 10^3 \times 0.671 \times 10^{-3}}{1 + 0.671 \times 10^{-3} \times 2 \times 10^3}$$

$$= - 1.22$$